BRCA-TCGA-APOLLO — foundation-model performance

Main classifiers — Accuracy

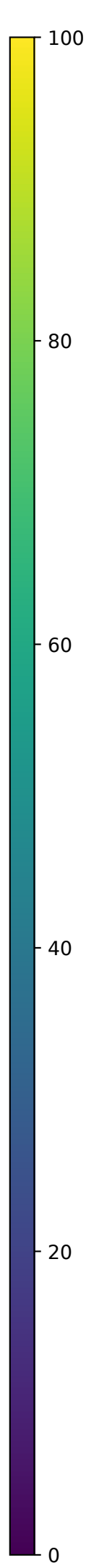
	Omiclass	Logistic Regression
DE baseline		
DE	74.1 ±4.2	N/A
DE · corrected	87.6 ±4.0	N/A
BulkFormer		
Full	75.0	75.0
Full · corrected	85.3	86.8
PCA	61.8	72.1
PCA · corrected	80.9	85.3
Supervised	76.5	72.1
Supervised · corrected	82.4	79.4
Supervised + PCA	69.1	73.5
Supervised + PCA · corrected	80.9	85.3
CellFM 1-view		
Full	57.4	54.4
Full · corrected	45.6	57.4
PCA	41.2	60.3
PCA · corrected	57.4	54.4
Supervised	42.6	52.9
Supervised · corrected	44.1	52.9
Supervised + PCA	54.4	50.0
Supervised + PCA · corrected	45.6	50.0
CellFM 20-view		
Full	79.4	75.0
Full · corrected	73.5	75.0
PCA	79.4	72.1
PCA · corrected	83.8	72.1
Supervised	70.6	66.2
Supervised · corrected	77.9	66.2
Supervised + PCA	72.1	66.2
Supervised + PCA · corrected	72.1	66.2
BulkRNABert		
Full	72.1	61.8
Full · corrected	72.1	67.6
PCA	55.9	61.8
PCA · corrected	72.1	58.8
Supervised	69.1	58.8
Supervised · corrected	69.1	57.4
Supervised + PCA	57.4	54.4
Supervised + PCA · corrected	64.7	55.9

Main classifiers — Macro-F1

	Omiclass	Logistic Regression
DE baseline		
DE	78.0 ±3.9	N/A
DE · corrected	84.9 ±5.6	N/A
BulkFormer		
Full	54.8	70.4
Full · corrected	80.6	80.9
PCA	38.5	68.5
PCA · corrected	73.1	79.6
Supervised	53.0	66.6
Supervised · corrected	77.2	72.0
Supervised + PCA	58.7	66.9
Supervised + PCA · corrected	73.0	79.9
CellFM 1-view		
Full	40.0	38.2
Full · corrected	28.0	45.0
PCA	35.6	55.5
PCA · corrected	50.0	48.7
Supervised	31.9	42.4
Supervised · corrected	30.8	41.5
Supervised + PCA	26.3	32.7
Supervised + PCA · corrected	32.5	38.8
CellFM 20-view		
Full	72.2	72.5
Full · corrected	67.3	72.5
PCA	63.1	70.2
PCA · corrected	68.3	68.9
Supervised	58.7	64.9
Supervised · corrected	53.9	63.4
Supervised + PCA	59.9	66.7
Supervised + PCA · corrected	52.7	63.3
BulkRNABert		
Full	68.8	61.3
Full · corrected	42.4	38.9
PCA	48.7	62.1
PCA · corrected	45.5	28.5
Supervised	68.4	58.4
Supervised · corrected	49.2	27.7
Supervised + PCA	40.5	56.9
Supervised + PCA · corrected	36.7	25.4

Additional classifiers — Accuracy

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	79.4	72.1	79.4
Full · corrected	85.3	80.9	83.8
PCA	75.0	69.1	60.3
PCA · corrected	86.8	86.8	83.8
Supervised	82.4	82.4	77.9
Supervised · corrected	82.4	79.4	89.7
Supervised + PCA	64.7	66.2	60.3
Supervised + PCA · corrected	80.9	80.9	77.9
Full tokens	80.9	N/A	N/A
Full tokens · corrected	82.4	N/A	N/A
CellFM 1-view			
Full	51.5	54.4	52.9
Full · corrected	58.8	55.9	52.9
PCA	58.8	52.9	58.8
PCA · corrected	50.0	54.4	58.8
Supervised	52.9	48.5	52.9
Supervised · corrected	51.5	51.5	52.9
Supervised + PCA	48.5	51.5	51.5
Supervised + PCA · corrected	47.1	51.5	51.5
Full tokens	45.6	N/A	N/A
Full tokens · corrected	45.6	N/A	N/A
CellFM 20-view			
Full	70.6	61.8	79.4
Full · corrected	66.2	48.5	79.4
PCA	79.4	75.0	79.4
PCA · corrected	75.0	77.9	80.9
Supervised	69.1	66.2	73.5
Supervised · corrected	72.1	67.6	73.5
Supervised + PCA	66.2	64.7	73.5
Supervised + PCA · corrected	69.1	66.2	69.1
Full tokens	79.4	N/A	N/A
Full tokens · corrected	73.5	N/A	N/A
BulkRNABert			
Full	64.7	72.1	76.5
Full · corrected	72.1	70.6	52.9
PCA	60.3	57.4	69.1
PCA · corrected	60.3	80.9	52.9
Supervised	63.2	69.1	75.0
Supervised · corrected	73.5	73.5	54.4
Supervised + PCA	60.3	60.3	75.0
Supervised + PCA · corrected	61.8	67.6	52.9

Additional classifiers — Macro-F1

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	76.5	66.4	57.4
Full · corrected	77.9	67.7	78.2
PCA	58.5	56.8	40.6
PCA · corrected	84.0	83.1	77.7
Supervised	77.9	68.7	59.9
Supervised · corrected	75.7	66.8	87.4
Supervised + PCA	40.0	36.8	37.7
Supervised + PCA · corrected	74.2	73.2	68.7
Full tokens	84.4	N/A	N/A
Full tokens · corrected	78.2	N/A	N/A
CellFM 1-view			
Full	34.4	36.7	17.3
Full · corrected	47.3	39.9	17.3
PCA	51.3	36.7	28.4
PCA · corrected	39.6	48.5	28.2
Supervised	40.1	41.0	25.6
Supervised · corrected	39.8	40.6	25.6
Supervised + PCA	30.6	32.4	29.2
Supervised + PCA · corrected	34.4	37.4	29.8
Full tokens	18.8	N/A	N/A
Full tokens · corrected	25.7	N/A	N/A
CellFM 20-view			
Full	52.8	41.8	64.4
Full · corrected	57.3	46.9	64.4
PCA	77.7	70.9	66.0
PCA · corrected	72.2	74.3	71.6
Supervised	63.3	62.8	66.5
Supervised · corrected	64.3	64.4	66.5
Supervised + PCA	58.0	57.2	54.0
Supervised + PCA · corrected	64.8	63.7	57.9
Full tokens	64.9	N/A	N/A
Full tokens · corrected	47.3	N/A	N/A
BulkRNABert			
Full	56.3	61.5	68.0
Full · corrected	42.5	40.7	17.3
PCA	58.9	56.5	47.0
PCA · corrected	30.9	59.6	17.3
Supervised	52.3	61.7	62.6
Supervised · corrected	46.1	43.1	20.6
Supervised + PCA	59.5	61.1	52.9
Supervised + PCA · corrected	33.4	44.8	17.3